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Application of single-step single nucleotide polymorphism best linear unbiased predictor model with unknown-parent groups for type traits in Japanese Holsteins

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ABSTRACT

The objectives of this study were to investigate the computational performance and the predictive ability and bias of a single-step SNP BLUP model (ssSNPBLUP) in genotyped young animals with unknown-parent groups (UPG) for type traits, using national genetic evaluation data from the Japanese Holstein population. The phenotype, genotype, and pedigree data were the same as those used in a national genetic evaluation of linear type traits classified between April 1984 and December 2020. In the current study, 2 data sets were prepared: the full data set containing all entries up to December 2020 and a truncated data set ending with December 2016. Genotyped animals were classified into 3 types: sires with classified daughters (S), cows with records (C), and young animals (Y). The computing performance and prediction accuracy of ssSNPBLUP were compared for the following 3 groups of genotyped animals: sires with classified daughters and young animals (SY); cows with records and young animals (CY); and sires with classified daughters, cows with records, and young animals (SCY). In addition, we tested 3 parameters of residual polygenic variance in ssSNPBLUP (0.1, 0.2, or 0.3). Daughter yield deviations (DYD) for the validation bulls and phenotypes adjusted for all fixed effects and random effects other than animal and residual (Y_{adj}) for the validation cows were obtained using the full data set from the pedigree-based BLUP model. The regression coefficients of DYD for bulls (or Y_{adj} for cows) on the genomic estimated breeding value (GEBV) using the truncated data set were used to measure the inflation of the predictions of young animals. The coefficient of determination of DYD on GEBV was used to measure the predictive ability of the predictions for the validation bulls. The reliability of the predictions for the validation cows was calculated as the square of the correlation between Y_{adj} and GEBV divided by heritability. The predictive ability was highest in the SCY group and lowest in the CY group. However, minimal difference was found in predictive abilities with or without UPG models using different parameters of residual polygenic variance. The regression coefficients approached 1.0 as the parameter of residual polygenic variance increased, but regression coefficients were mostly similar regardless of the use of UPG across the groups of genotyped animals. The ssSNPBLUP model, including UPG, was demonstrated as feasible for implementation in the national evaluation of type traits in Japanese Holsteins.

Key words: single-step genomic evaluation, single nucleotide polymorphism best linear unbiased predictor, unknown-parent groups, type trait, Holstein

INTRODUCTION

Currently, genomic evaluation systems for dairy cattle have been implemented in many countries using large-scale SNP genotypes. For this purpose, most countries have applied a multistep method (VanRaden, 2008) that typically uses deregressed EBV as a pseudo-phenotype obtained via pedigree-based genetic evaluations. This method requires a set of genotyped animals, referred to as a reference population, that have reliable pseudo-phenotypes to estimate SNP effects or genomic EBV (**GEBV**). In this approach, genomic prediction is an extra step added to a conventional genetic evaluation. One possible disadvantage of the multistep method is a preselection bias in conventional EBV (Patry and Ducrocq, 2011). When young animals are selected for many generations based on GEBV from a multistep model, the conventional EBV will be biased because genomic preselection is not accounted for in the conventional evaluation.

The single-step genomic BLUP model (**ssGBLUP**) provides GEBV for genotyped and nongenotyped animals (Aguilar et al., 2010), and it can account for genomic preselection in young animals; therefore, the

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GEBV is expected to be unbiased (Masuda et al., 2018). This model integrates the genomic relationship matrix **G** and pedigree relationship matrix **A** into the hybrid matrix **H** and replaces the matrix **A** in the BLUP equations. The matrix **H** may not be easily determined in an actual population. Because the scales of **G** and **A** may not be identical, **G** should be aligned to the relationship matrix for genotyped animals, $\mathbf{A}_{22}$, or vice versa, using scaling factors. In addition, when including a substantial number of genotyped animals, computing costs will be very high for calculating the inverse of **G**, which is required in the mixed model equations in ssGBLUP. Therefore, efficient techniques such as the algorithm for proven and young animals (Misztal et al., 2014; Fragomeni et al., 2015) should be used to avoid the high computing costs required for large-scale genotype data, such as a national-level database.

Liu et al. (2014, 2016) developed a single-step SNP BLUP model (**ssSNPBLUP**) that is theoretically equivalent to ssGBLUP. It does not require **G** and its inverse, and it can be relatively easily introduced into computer programs to solve traditional BLUP equations. Initially, most studies focused on ssGBLUP, but recently, studies have increasingly focused on ssSNPBLUP. Osawa and Saburi (2018) applied this method to a repeatability model for 305-d milk yields in Japanese Holstein cows with approximately 5,500 genotyped animals. Konstantinov and Goddard (2020) applied the method to multivariate models, that is, a random regression model and a multiple trait model in Australian dairy cattle. Alkhoder et al. (2022) compared ssSNPBLUP and a routine multistep model for conformation traits in German Holsteins and demonstrated the superiority of ssSNPBLUP.

Generally, ssSNPBLUP is reported to have poorer convergence than ssGBLUP (Taskinen et al., 2017) when the preconditioned conjugate gradient (**PCG**) method is used. Vandenplas et al. (2018, 2019) developed special preconditioners for SNP effects in the PCG method to be applied for large-scale genomic evaluation.

The national genetic evaluation of dairy cattle usually considers unknown-parent groups (**UPG**), known as phantom parents or genetic groups, to compensate for past selections that cannot be accounted for by incomplete pedigree relationships with missing parents. The UPG model for pedigree-based BLUP (Quaas, 1988) was extended by Misztal et al. (2013), who applied the Quaas and Pollak (1981) (**QP**) transformation to ssGBLUP. Vandenplas et al. (2021) proposed the application of QP transformation to ssSNPBLUP, but this method has not been tested thoroughly in the Japanese dairy populations.

Therefore, we investigated the computational performance and the predictive ability and bias of the ssSNPBLUP model with UPG for type traits of genotyped young animals, using national genetic evaluation data from the Japanese Holstein population.

MATERIALS AND METHODS

Data

The phenotype and pedigree data used in this study were the same as those used in the national genetic evaluation of linear type traits for Japanese Holsteins, as carried out by the National Livestock Breeding Center (**NLBC**, Nishigo-mura, Fukushima, Japan). The phenotypes of the first 3 lactations were provided by the Holstein Cattle Association of Japan (**HCAJ**, Nakano-ku, Tokyo). Cows were classified between April 1984 and December 2020. In this study, 2 data sets were prepared to validate the predictive ability of ssSNPBLUP. One data set contained records for type traits of the full data set up to December 2020, and the other was the truncated data set with the type traits ending with December 2016. The number of records and animals in the full and truncated data sets are shown in Table 1. Thirteen linear type traits (stature, chest width, body depth, angularity, rump angle, rear leg side view, foot angle, fore udder attachment, rear udder height, rear udder width, udder support, udder depth, and front teat placement) were analyzed in this study. All traits were scored on a 9-point scale by HCAJ classifiers and adjusted for heterogeneous variance in a contemporary group (Strabel et al., 2006). The heritability and repeatability of these 13 traits are presented in Table 2.

Table 1. Number of records and animals in the full and truncated data sets

Item	Full data set	Truncated data set
Number of records	1,880,607	1,687,715
First parity	1,248,182	1,106,504
Second parity	412,428	379,847
Third parity	219,997	201,364
Pedigree	2,810,212	2,440,784
Number of cows with records	1,476,987	1,325,097
Number of sires with classified daughters	10,706	9,464

Table 2. Heritabilities and repeatabilities of linear type traits

Trait	Heritability	Repeatability
Stature	0.51	0.66
Chest width	0.26	0.39
Body depth	0.35	0.51
Angularity	0.20	0.30
Rump angle	0.42	0.64
Rear leg side view	0.21	0.36
Foot angle	0.06	0.13
Fore udder attachment	0.22	0.40
Rear udder height	0.27	0.45
Rear udder width	0.21	0.35
Udder support	0.20	0.36
Udder depth	0.46	0.62
Front teat placement	0.39	0.52

The study included a total of 109,766 genotyped animals and the details are shown in Table 3. Genotyped animals were divided into 2 groups based on the date of birth. One group included the full data set of animals born in November 2020 or earlier, and the other comprised the truncated data set with animals born in November 2016 or earlier. Most sires with classified daughters were genotyped with a medium-density SNP chip equivalent to the Illumina BovineSNP 50K array, whereas most of the cows with records and the young animals were genotyped with a low-density SNP chip equivalent to the Illumina BovineSNP 7K array. Animals with lower density genotypes were imputed to 50K. Genotype imputing was performed with a program developed in house based on the method of VanRaden et al. (2011). Animals with call rates of <0.90 were excluded, and the number of SNP markers used in this study was 42,275, which was the same used in routine national genomic evaluation by NLBC in Japan.

Genotyping Scenario

Three groups were defined for genotyped animals: sires with classified daughters (**S**), cows with records (**C**), and young animals (**Y**). Note that the definition of young animals in this analysis included males without own daughter 7 yr of age or less and females without own record less than 4 yr of age or less. The numbers of genotyped animals are shown in Table 3. We created 3 data sets, each with identical phenotypes and pedigrees but with a different group of genotyped animals. Then, ssSNPBLUP was applied to each data set, and we compared computing performance and prediction accuracy among the 3 groups (scenarios). The first group consisted of sires with classified daughters and young animals (**SY**), the second group consisted of cows with records and young animals (**CY**), and the third group consisted of sires with classified daughters, cows with records, and young animals (**SCY**; i.e., all genotyped animals). The numbers of genotyped animals of SY, CY, and SCY were 68,209, 104,086, and 109,766 in the full data set, and 47,367, 49,613, and 54,264 in the truncated data set, respectively (Table 3). The trend in the number of genotyped animals is shown in Figure 1. Most cows had been genotyped in the preceding 5 years, resulting in a significant reduction in the number of genotyped cows with records in the truncated data set up to 2016.

Single-Step SNP BLUP Model

The genomic model for type traits was similar to the single-trait repeatability model used in routine national genetic evaluation by NLBC in Japan (http://www.nlbc.go.jp/en/sec03.html).

$$\mathbf{y} = \mathbf{Xb} + \mathbf{Z}_p\mathbf{p} + \mathbf{Wu} + \mathbf{e}, \quad [1]$$

where $\mathbf{y}$ is the vector of observations (pre-adjusted for heterogeneous variances), $\mathbf{b}$ is the vector of fixed effects, $\mathbf{p}$ is the vector of permanent environmental effects, $\mathbf{u}$ is the vector of additive genetic effects, and $\mathbf{e}$ is the vector of random residuals. The matrices $\mathbf{X}$, $\mathbf{Z}_p$, and $\mathbf{W}$ are the incidence matrices for $\mathbf{b}$, $\mathbf{p}$, and $\mathbf{u}$, respectively. An

Table 3. Number of genotyped animals with different groups of genotyped animals (SY, CY, and SCY) in the full and truncated data sets

Item	Full data set	Truncated data set
Number of genotyped animals		
Sire with classified daughters (S)	5,680	4,651
Cow with records (C)	41,557	6,897
Young animal (Y)	62,529	42,716
Male	5,930	4,032
Female	56,599	38,684
Genotyping scenario		
Sire with classified daughters and young animal (SY)	68,209	47,367
Cow with records and young animal (CY)	104,086	49,613
Sire with classified daughters, cow with records, and young animal (SCY)	109,766	54,264

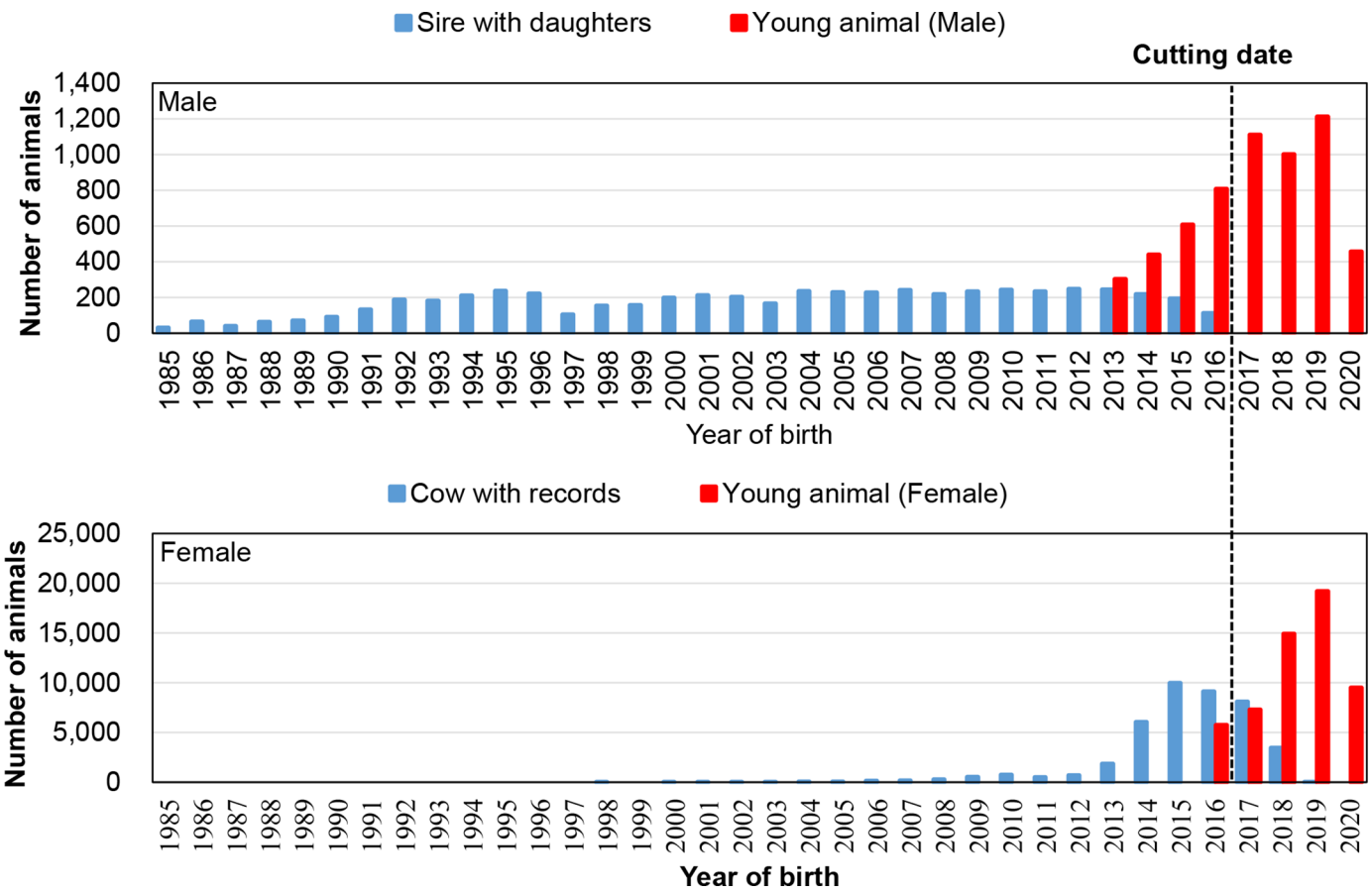


Figure 1. The number of genotyped animals (sires with classified daughters, cows with records, and young animals) by birth year. The dotted line indicates the cutting date, the last day of truncated data.

additional fixed covariate, the so-called J factor (Hsu et al., 2017), was included in ssSNPBLUP. The purpose of including it was to correct for the difference of genetic levels between the genotyped and nongenotyped animals when genotypes were centered using the observed allele frequencies instead of the base population allele frequencies. The system of equations for ssSNPBLUP can be written as follows:

$$\begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{Z}_p & \mathbf{X}'\mathbf{R}^{-1}\mathbf{W}_1 & \mathbf{X}'\mathbf{R}^{-1}\mathbf{W}_2 & \mathbf{0} \\ \mathbf{Z}_p'\mathbf{R}^{-1}\mathbf{X} & \mathbf{Z}_p'\mathbf{R}^{-1}\mathbf{Z}_p + \sigma_p^{-2}\mathbf{I} & \mathbf{Z}_p'\mathbf{R}^{-1}\mathbf{W}_1 & \mathbf{Z}_p'\mathbf{R}^{-1}\mathbf{W}_2 & \mathbf{0} \\ \mathbf{W}_1'\mathbf{R}^{-1}\mathbf{X} & \mathbf{W}_1'\mathbf{R}^{-1}\mathbf{Z}_p & \mathbf{W}_1'\mathbf{R}^{-1}\mathbf{W}_1 + \sigma_u^{-2}\boldsymbol{\Sigma}^{11} & \sigma_u^{-2}\boldsymbol{\Sigma}^{12} & \sigma_u^{-2}\boldsymbol{\Sigma}^{1g} \\ \mathbf{W}_2'\mathbf{R}^{-1}\mathbf{X} & \mathbf{W}_2'\mathbf{R}^{-1}\mathbf{Z}_p & \sigma_u^{-2}\boldsymbol{\Sigma}^{21} & \mathbf{W}_2'\mathbf{R}^{-1}\mathbf{W}_2 + \sigma_u^{-2}\boldsymbol{\Sigma}^{22} & \sigma_u^{-2}\boldsymbol{\Sigma}^{2g} \\ \mathbf{0} & \mathbf{0} & \sigma_u^{-2}\boldsymbol{\Sigma}^{g1} & \sigma_u^{-2}\boldsymbol{\Sigma}^{g2} & \sigma_u^{-2}\boldsymbol{\Sigma}^{gg} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{b}} \\ \hat{\mathbf{p}} \\ \hat{\mathbf{u}}_1 \\ \hat{\mathbf{u}}_2 \\ \hat{\mathbf{g}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}_p'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{W}_1'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{W}_2'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{0} \end{bmatrix} \quad [2]$$

with

$$\boldsymbol{\Sigma}^{-1} = \begin{bmatrix} \boldsymbol{\Sigma}^{11} & \boldsymbol{\Sigma}^{12} & \boldsymbol{\Sigma}^{1g} \\ \boldsymbol{\Sigma}^{21} & \boldsymbol{\Sigma}^{22} & \boldsymbol{\Sigma}^{2g} \\ \boldsymbol{\Sigma}^{g1} & \boldsymbol{\Sigma}^{g2} & \boldsymbol{\Sigma}^{gg} \end{bmatrix} = \begin{bmatrix} \mathbf{A}^{11} & \mathbf{A}^{12} & 0 \\ \mathbf{A}^{21} & \mathbf{A}^{22} + \left(\frac{1}{k} - 1\right)\mathbf{A}_{22}^{-1} & -\frac{1}{k}\mathbf{A}_{22}^{-1}\mathbf{Z} \\ 0 & -\frac{1}{k}\mathbf{Z}'\mathbf{A}_{22}^{-1} & \frac{1}{1-k}\mathbf{B}^{-1} + \frac{1}{k}\mathbf{Z}'\mathbf{A}_{22}^{-1}\mathbf{Z} \end{bmatrix}$$

and

$$\mathbf{A}^{-1} = \begin{bmatrix} \mathbf{A}^{11} & \mathbf{A}^{12} \\ \mathbf{A}^{21} & \mathbf{A}^{22} \end{bmatrix},$$

where superscripts 1, 2, and g refer to groups of nongenotyped animals, genotyped animals, and SNP effects, respectively, $\hat{\mathbf{u}}_1$ is the vector of additive genetic effects of nongenotyped animals, $\hat{\mathbf{u}}_2$ is a vector of the additive genetic effects of genotyped animals, $\hat{\mathbf{g}}$ is a vector of SNP effects, $\mathbf{W}_1$ and $\mathbf{W}_2$ are incidence matrices for the additive genetic effects of nongenotyped and genotyped animals, respectively, $\mathbf{R} = \sigma_e^2\mathbf{I}$, and $\mathbf{A}^{11}$, $\mathbf{A}^{12}$, $\mathbf{A}^{21}$, and $\mathbf{A}^{22}$ are submatrices of the inverse of the pedigree relationship matrix $\mathbf{A}^{-1}$ for all animals, $\mathbf{A}_{22}^{-1}$ is the inverse of the pedigree relationship matrix for genotyped animals ($\mathbf{A}_{22}$). $\mathbf{Z}$ is a design matrix of regression coefficients for genotyped animals at all SNP markers [$2 - 2p_j$, $1 - 2p_j$, or $-2p_j$, for genotypes AA, AB, or BB of the jth SNP marker, respectively, as in VanRaden (2008), where p_j represents the observed allele frequency of the jth SNP marker]; $\mathbf{B}$ is $\left[\sum_{j=1}^{m} 2p_j\left(1-p_j\right)\right]^{-1}\mathbf{I}$; and σ_p^2, σ_u^2, and σ_e^2 are permanent environmental, additive genetic, and residual variances, respectively. The parameter of residual polygenic variance (k) ranges from 0 to 1, and 3 values (0.1, 0.2, and 0.3) were tested in this study. Liu et al. (2014) presented a calculation procedure that separated the estimation process of SNP effects from Equation [2]; this approach was used by Osawa and Saburi (2018). However, we solved Equation [2] without separating it, as in Konstantinov and Goddard (2020).

Implementation of Genetic Groups in the ssSNPBLUP Model

Vandenplas et al. (2021) applied a QP transformation (Quaas and Pollak, 1981) to ssSNPBLUP to incorporate UPG into Equation [2]. This method is equivalent to the system of equations derived by Misztal et al. (2013) for ssGBLUP. The QP-transformed system of Equation [2] is as follows:

$$\begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{Z}_p & \mathbf{X}'\mathbf{R}^{-1}\mathbf{W}_1 & \mathbf{X}'\mathbf{R}^{-1}\mathbf{W}_2 & \mathbf{0} & \mathbf{0} \\ \mathbf{Z}_p'\mathbf{R}^{-1}\mathbf{X} & \mathbf{Z}_p'\mathbf{R}^{-1}\mathbf{Z}_p + \sigma_p^{-2}\mathbf{I} & \mathbf{Z}_p'\mathbf{R}^{-1}\mathbf{W}_1 & \mathbf{Z}_p'\mathbf{R}^{-1}\mathbf{W}_2 & \mathbf{0} & \mathbf{0} \\ \mathbf{W}_1'\mathbf{R}^{-1}\mathbf{X} & \mathbf{W}_1'\mathbf{R}^{-1}\mathbf{Z}_p & \mathbf{W}_1'\mathbf{R}^{-1}\mathbf{W}_1 + \sigma_u^{-2}\boldsymbol{\Sigma}_{QP}^{11} & \sigma_u^2\boldsymbol{\Sigma}_{QP}^{12} & \sigma_u^{-2}\boldsymbol{\Sigma}_{QP}^{1t} & \sigma_u^{-2}\boldsymbol{\Sigma}_{QP}^{1g} \\ \mathbf{W}_2'\mathbf{R}^{-1}\mathbf{X} & \mathbf{W}_2'\mathbf{R}^{-1}\mathbf{Z}_p & \sigma_u^{-2}\boldsymbol{\Sigma}_{QP}^{21} & \mathbf{W}_2'\mathbf{R}^{-1}\mathbf{W}_2 + \sigma_u^2\boldsymbol{\Sigma}_{QP}^{22} & \sigma_u^{-2}\boldsymbol{\Sigma}_{QP}^{2t} & \sigma_u^{-2}\boldsymbol{\Sigma}_{QP}^{2g} \\ \mathbf{0} & \mathbf{0} & \sigma_u^{-2}\boldsymbol{\Sigma}_{QP}^{t1} & \sigma_u^2\boldsymbol{\Sigma}_{QP}^{t2} & \sigma_u^{-2}\boldsymbol{\Sigma}_{QP}^{tt} & \sigma_u^{-2}\boldsymbol{\Sigma}_{QP}^{tg} \\ \mathbf{0} & \mathbf{0} & \sigma_u^{-2}\boldsymbol{\Sigma}_{QP}^{g1} & \sigma_u^2\boldsymbol{\Sigma}_{QP}^{g2} & \sigma_u^{-2}\boldsymbol{\Sigma}_{QP}^{gt} & \sigma_u^{-2}\boldsymbol{\Sigma}_{QP}^{gg} \end{bmatrix} \quad [3]$$

$$\begin{bmatrix} \hat{\mathbf{b}} \\ \hat{\mathbf{p}} \\ \hat{\mathbf{a}}_1 \\ \hat{\mathbf{a}}_2 \\ \hat{\mathbf{t}} \\ \hat{\mathbf{g}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}_p'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{W}_1'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{W}_2'\mathbf{R}^{-1}\mathbf{y} \\ 0 \\ 0 \end{bmatrix}$$

with

$$\mathbf{Q} = \begin{bmatrix} \mathbf{Q}_1 \\ \mathbf{Q}_2 \end{bmatrix},$$

$$\hat{\mathbf{a}}_1 = \mathbf{Q}_1\hat{\mathbf{t}} + \hat{\mathbf{u}}_1,$$

$$\hat{\mathbf{a}}_2 = \mathbf{Q}_2\hat{\mathbf{t}} + \hat{\mathbf{u}}_2,$$

and

$$\boldsymbol{\Sigma}_{QP}^{-1} = \begin{bmatrix} \boldsymbol{\Sigma}_{QP}^{11} & \boldsymbol{\Sigma}_{QP}^{12} & \boldsymbol{\Sigma}_{QP}^{1t} & \boldsymbol{\Sigma}_{QP}^{1g} \\ \boldsymbol{\Sigma}_{QP}^{21} & \boldsymbol{\Sigma}_{QP}^{22} & \boldsymbol{\Sigma}_{QP}^{2t} & \boldsymbol{\Sigma}_{QP}^{2g} \\ \boldsymbol{\Sigma}_{QP}^{t1} & \boldsymbol{\Sigma}_{QP}^{t2} & \boldsymbol{\Sigma}_{QP}^{tt} & \boldsymbol{\Sigma}_{QP}^{tg} \\ \boldsymbol{\Sigma}_{QP}^{g1} & \boldsymbol{\Sigma}_{QP}^{g2} & \boldsymbol{\Sigma}_{QP}^{gt} & \boldsymbol{\Sigma}_{QP}^{gg} \end{bmatrix}$$

$$= \begin{bmatrix} \mathbf{A}^{11} & \mathbf{A}^{12} & -\mathbf{A}^{11}\mathbf{Q}_1 - \mathbf{A}^{12}\mathbf{Q}_1 & 0 \\ \mathbf{A}^{21} & \mathbf{A}^{22} & -\mathbf{A}^{21}\mathbf{Q}_1 - \mathbf{A}^{22}\mathbf{Q}_2 & 0 \\ -\mathbf{Q}_1'\mathbf{A}^{11} - \mathbf{Q}_2'\mathbf{A}^{21} & -\mathbf{Q}_1'\mathbf{A}^{12} - \mathbf{Q}_2'\mathbf{A}^{22} & \mathbf{Q}'\mathbf{A}^{-1}\mathbf{Q} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} +$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & \left(\frac{1}{k}-1\right)\mathbf{A}_{22}^{-1} & -\left(\frac{1}{k}-1\right)\mathbf{A}_{22}^{-1}\mathbf{Q}_2 & -\frac{1}{k}\mathbf{A}_{22}^{-1}\mathbf{Z} \\ 0 & -\left(\frac{1}{k}-1\right)\mathbf{Q}_2'\mathbf{A}_{22}^{-1} & \left(\frac{1}{k}-1\right)\mathbf{Q}_2'\mathbf{A}_{22}^{-1}\mathbf{Q}_2 & \frac{1}{k}\mathbf{Q}_2'\mathbf{A}_{22}^{-1}\mathbf{Z} \\ 0 & -\frac{1}{k}\mathbf{Z}'\mathbf{A}_{22}^{-1} & \frac{1}{k}\mathbf{Z}'\mathbf{A}_{22}^{-1}\mathbf{Q}_2 & \frac{1}{1-\mathrm{k}}\mathbf{B}^{-1} + \frac{1}{\mathrm{k}}\mathbf{Z}'\mathbf{A}_{22}^{-1}\mathbf{Z} \end{bmatrix},$$

where superscript t refers to genetic groups, $\hat{\mathbf{t}}$ is a vector of genetic group effects, and $\mathbf{Q}$ is a matrix that assigns animals to genetic groups. In this study, 81 genetic groups were defined based on the birth year, sex, and selection path used in the national routine genetic evaluation in Japan. The orders of $\mathbf{Q}_2'\mathbf{A}_{22}^{-1}\mathbf{Q}_2$ and $\mathbf{Q}_2'\mathbf{A}_{22}^{-1}\mathbf{Z}$ in all genotyping scenarios were 81 by 81 and 81 by 42,275, respectively, and the order of $\mathbf{A}_{22}^{-1}\mathbf{Q}_2$ was 68,209 by 81, 104,086 by 81, and 109,766 by 81 in SY, CY, and SCY, respectively.

Preconditioned Conjugate Gradient Method and Preconditioner

We also adopted the PCG to solve the above equations. A solver supporting the ssSNPBLUP model was developed to process large amounts of data in routine national genetic evaluations. The computer program used the iteration on data algorithm (Lidauer et al., 1999; Tsuruta et al., 2001). Before running PCG iterations, we first calculated $\mathbf{A}_{22}^{-1}\mathbf{Z}$, $\frac{1}{1-k}\mathbf{B}^{-1} + \frac{1}{k}\mathbf{Z}'\mathbf{A}_{22}^{-1}\mathbf{Z}$, $\mathbf{A}_{22}^{-1}\mathbf{Q}_2$, $\mathbf{Q}_2'\mathbf{A}_{22}^{-1}\mathbf{Q}_2$, and $\mathbf{Q}_2'\mathbf{A}_{22}^{-1}\mathbf{Z}$, using indirect multiplication algorithms of $\mathbf{A}_{22}^{-1}\mathbf{Q}$ that do not require calculations of the inverse $\mathbf{A}_{22}^{-1}$ (Masuda et al., 2017; Strandén et al., 2017). The matrix products were stored as double precision real numbers in random access memory (**RAM**). This program was written in Fortran 90/95 and compiled using the Intel Fortran Compiler 2019 (Intel Corporation) with the Basic Linear Algebra Subprograms (BLAS), Linear Algebra Package (LAPACK), and direct sparse solver PARDISO provided by the Intel Math Kernel Library 2019. Parallel computing using OpenMP was implemented in this program. Eight OpenMP threads were used for all the computations. All analyses were performed on a Linux server with 768 GB of RAM and 2 Intel Xeon Gold 6252 (2.10 GHz) processors with 48 physical cores in total.

The system of Equations [2] or [3], which is written as $\mathbf{Cx} = \mathbf{b}$, can be written as $\mathbf{M}^{-1}\mathbf{Cx} = \mathbf{M}^{-1}\mathbf{b}$ with the preconditioner matrix $\mathbf{M}$ in the PCG. Specifically, $\mathbf{M}$ causes the condition number of the left-hand side matrix to be small and improves the convergence. As a result of the preanalysis, the preconditioner matrix $\mathbf{M}$ was chosen, as suggested by Konstantinov and Goddard (2020):

$$\mathbf{C} = \begin{bmatrix} \mathbf{C}_{oo} & \mathbf{C}_{og} \\ \mathbf{C}_{go} & \mathbf{C}_{gg} \end{bmatrix},$$

$$\mathbf{M} = \begin{bmatrix} \mathrm{diag}\left(\mathbf{C}_{oo}\right) & 0 \\ 0 & \mathbf{C}_{gg} \end{bmatrix},$$

and

$$\mathbf{C}_{gg} = \sigma_a^{-2}\left(\frac{1}{1-k}\mathbf{B}^{-1} + \frac{1}{k}\mathbf{Z}'\mathbf{A}_{22}^{-1}\mathbf{Z}\right),$$

where $\mathbf{C}$ is the left-hand side matrix in Equations [2] or [3], and subscripts g and o refer to parts of the SNP effects and other effects, respectively. In this study, the convergence criterion was defined as $||\,\mathbf{b} - \mathbf{Cx}\,||^2 \,/\, ||\,\mathbf{b}\,||^2 < 10^{-14}$, where $||\cdot||$ is the 2-norm.

Validation

Three types of GEBV (SY, CY, and SCY) were calculated for each trait. In addition, the EBV obtained using pedigree-based BLUP with the truncated data set was calculated and used as a control.

To validate GEBV, we defined the validation bulls as genotyped young bulls that had no daughters in the truncated data set but had at least 15 daughters in 10 herds in the full data set. The validation cows were genotyped heifers in the truncated data set, but had their own records in the full data set. Daughter yield deviations (**DYD**) for the validation bulls as well as phenotypes adjusted for all fixed and random effects other than animal and residual (Y_{adj}) for the validation cows were obtained using the full data set from the pedigree-based BLUP model with UPG. Note than Y_{adj} for cows with multiple records used the average value for each animal. In addition, DYD were estimated using the method described by VanRaden and Wiggans (1991).

The regression coefficients of DYD for bulls (or Y_{adj} for cows) based on EBV or GEBV were used to measure the inflation of the predictions of young animals. The coefficient of determination of DYD for EBV or GEBV was used to measure the reliability of the predictions for the validation bulls. The reliability of the predictions for the validation cows was calculated as the square of the correlation between Y_{adj} and GEBV divided by heritability.

RESULTS AND DISCUSSION

Computational Performance and Convergence

Table 4 shows an overview of the convergence in the pedigree-based BLUP and ssSNPBLUP models for the 13 type traits. This shows a result with the parameter of residual polygenic variance = 0.1. As this parameter increased, the number of iterations decreased. The peak memory requirement was increased from 0.8 GB in pedigree-based BLUP to 62 to 98 GB in ssSNPBLUP, and the memory requirement in the iteration phase was also increased from 0.8 GB in pedigree-based BLUP to 51 to 65 GB in ssSNPBLUP due to the need to create and store matrices related to SNP markers of genotyped animals. The ssSNPBLUP model also required more iterations and time per iterations owing to more complex equations than pedigree-based BLUP. Nevertheless, the computing time was still acceptable for routine national genetic evaluation. Similar to the results of Vandenplas et al. (2021), implementing UPG via QP transformation slightly increased the calculation time. The SY group (sires with classified daughters and young animals) with the lowest number of genotyped animals (n = 68,209) had the shortest calculation time (approximately 4 s per iteration), whereas the SCY group (sires with classified daughters, cows with records, and young animals) with the highest number of genotyped animals (n = 109,766) had a slightly longer calculation time (approximately 5 s per iteration round). In contrast, for the CY group (cows with records and young animals) with a smaller number of genotyped animals (n = 104,086) than SCY, the calculation time (approximately 13 s per iteration round) increased significantly. This was due to the number of nonzero elements of $\mathbf{A}^{00}$ (superscript 0 refers to nongenotyped ancestors of genotyped animals) in indirect multiplication of $\mathbf{A}_{22}^{-1}\mathbf{Q}$ (Masuda et al., 2017) for CY was much larger than that in the other groups. The numbers of nonzero elements of $\mathbf{A}^{00}$ in SY, CY, and SCY were 645,866, 1,443,820, and 907,914, respectively. In CY, genotyped sires were treated as nongenotyped animals, and their ancestors added more nonzero elements to $\mathbf{A}^{00}$ than SY and SCY. As a result, the computational cost of $\left(\mathbf{A}^{00}\right)^{-1}\mathbf{x}=\mathbf{y}$ with PARDISO was estimated to have increased more in CY than

Table 4. Computational performance in pedigree-based BLUP and single-step (ss)SNPBLUP models (residual polygenic variance = 0.10)

Group[1]	UPG[2]	Number of iterations[3]	Time per iteration[4] (s)	Calculation time[3] (s)	Peak memory[5] (GB)	Memory in iterative phase[6] (GB)
PED	No	148–377	0.3	67–134	0.8	0.8
PED	Yes	188–321	0.4	84–123	0.8	0.8
SY	No	545–806	3.8	3,967–5,398	62	51
SY	Yes	342–1,007	3.9	3,569–5,491	64	52
CY	No	470–807	12.9	10,299–14,471	93	63
CY	Yes	446–980	13.1	10,537–16,471	95	63
SCY	No	492–839	5.4	5,802–7,305	96	65
SCY	Yes	459–1,027	5.6	5,673–10,018	98	65

[1]PED = pedigree-based BLUP; SY = ssSNPBLUP consisting of sires with classified daughters and young animals; CY = ssSNPBLUP consisting of cows with records and young animals; SCY = ssSNPBLUP consisting of sires with classified daughters, cows with records, and young animals.
[2]UPG = unknown-parent group.
[3]Range of 13 traits.
[4]Wall-clock time. Average of 13 traits.
[5]Peak memory is defined as peak resident set size ("high water mark," VmHWM) obtained from the Linux/proc virtual file system.
[6]Memory in iterative phase is defined as the amount of memory mapped in RAM (VmRSS) obtained from the Linux/proc virtual file system.

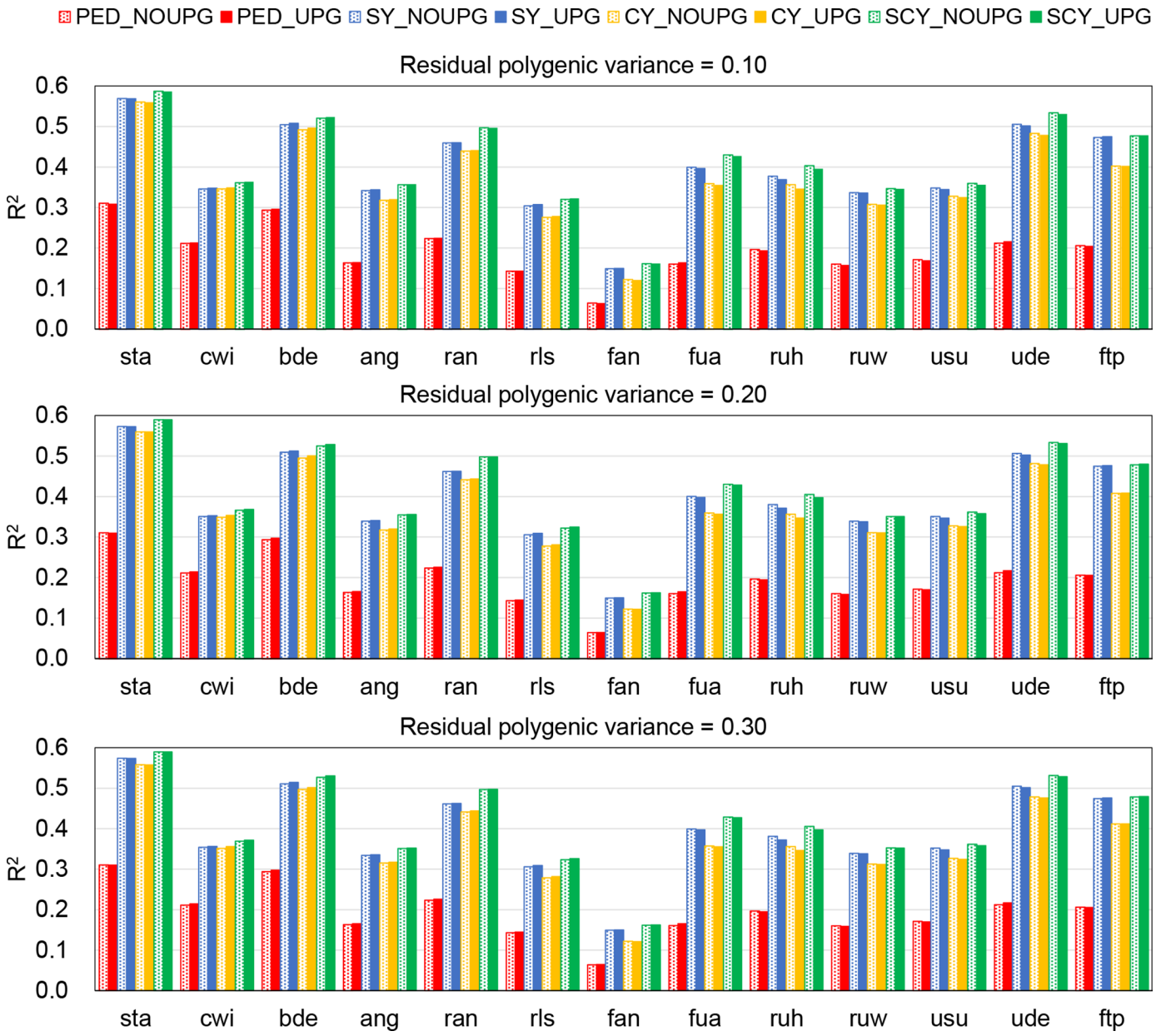


Figure 2. Coefficients of determination (R^2) for the validation bulls (n = 764) between daughter yield deviations (DYD) from pedigree-based BLUP in the full data set and EBV from pedigree-based BLUP (PED) as well as genomic estimated breeding value from single-step (ss) SNPBLUP by groups of genotyped animals in the truncated data set. PED_NOUPG and PED_UPG = pedigree-based BLUP without the unknown-parent group (UGP) and with UPG; SY_NOUPG and SY_UPG = ssSNPBLUP consisting of sires with classified daughters and young animals without UGP and with UPG; CY_NOUPG and CY_UPG = ssSNPBLUP consisting of cows with records and young animals without UGP and with UPG; SCY_NOUPG and SCY_UPG = ssSNPBLUP consisting of sires with classified daughters, cows with records, and young animals without UGP and with UPG; sta = stature; cwi = chest width; bde = body depth; ang = angularity; ran = rump angle; rls = rear leg side view; fan = foot angle; fua = fore udder attachment; ruh = rear udder height; ruw = rear udder width; usu = udder support; ude = udder depth; ftp = front teat placement.

SCY. This outcome is not an issue in the real-world genomic evaluation in Japan because CY was considered only as a scenario of comparison in this study; this group will not be used in practice. In our program, memory requirements were higher than in the study by Vandenplas et al. (2020) because the $\mathbf{A}_{22}^{-1}\mathbf{Z}$, $\frac{1}{1-k}\mathbf{B}^{-1}+\frac{1}{k}\mathbf{Z}'\mathbf{A}_{22}^{-1}\mathbf{Z}$, $\mathbf{A}_{22}^{-1}\mathbf{Q}_2$, $\mathbf{Q}_2'\mathbf{A}_{22}^{-1}\mathbf{Q}_2$, and $\mathbf{Q}_2'\mathbf{A}_{22}^{-1}\mathbf{Z}$ equations were dense and stored as double precision real numbers in the memory. $\mathbf{A}_{22}^{-1}\mathbf{Z}$ required the most memory, with 21.5 GB (size of 68,209 by 42,275), 32.8 GB (size of 104,086 by 42,275), and 34.6 GB (size of

109,766 by 42,275) in SY, CY, and SCY, respectively. The next highest was $\frac{1}{1-k}\mathbf{B}^{-1} + \frac{1}{k}\mathbf{Z}'\mathbf{A}_{22}^{-1}\mathbf{Z}$, with 13.3 GB (size of 42,275 by 42,275) for all analyses. Matrices related to genetic groups, $\mathbf{A}_{22}^{-1}\mathbf{Q}_2$, $\mathbf{Q}_2'\mathbf{A}_{22}^{-1}\mathbf{Q}_2$, and $\mathbf{Q}_2'\mathbf{A}_{22}^{-1}\mathbf{Z}$, were less than 1 GB. Peak memory was reached during the construction of $\mathbf{A}_{22}^{-1}\mathbf{Z}$, but memory decreased during the iterative phase. For example, in SCY, the peak memory was approximately 100 GB, and the memory in the iterative phase was reduced to 68 GB. These memory requirements were acceptable for routine national evaluations in Japan. Our program has not been fully optimized, and room remains for improvement. Implementing a compressed form (e.g., the Plink 1 binary form) for storing the genotyped matrix in RAM and computational strategies for the multiplication of matrix by a vector such as $\mathbf{A}_{22}^{-1}\mathbf{Z}\,\mathbf{x} = \mathbf{y}$ on the fly developed by Vandenplas et al. (2020) is expected to reduce the required RAM and improve computational efficiency.

Predictive Abilities of Validation Bulls and Cows

The coefficients of determination (R^2) of DYD for EBV [pedigree-based BLUP (**PED**)] or GEBV (SY, CY, and SCY) are shown in Figure 2. The R^2 of EBV from the PED was low (0.06–0.31), whereas the R^2 of GEBV from ssSNPBLUP was higher than that of PED by an average of 0.20. The SCY group was the most accurate and the CY group was the least accurate. This result likely occurred because the number of genotyped cows with records in the truncated data set was similar to the number of genotyped sires with classified daughters, whereas the reliability of cows with records was lower than that of proven sires. Moreover, not much difference was found in the predictive abilities between the models with and without UPG using different parameters of residual polygenic variance. In Japan, Holsteins must have pedigree registration records before the animals are genotyped. Thus, most genotyped animals have known parents. In addition, only pedigree registered cows have type traits. These factors may account for little differences between evaluations with and without UPG.

Correlations and reliabilities of predictions for cows from PED and ssSNPBLUP based on the 3 groups of genotyped animals are shown in Table 5. Similar to the predictive abilities for the validation bulls, no difference was found in the predictive abilities of the validation cows owing to differences in UPG and residual polygenic variance; only the results with UPG are shown. The average (ranges) of correlations across traits were 0.31 (0.10–0.50) for PED, 0.39 (0.13–0.60) for SY, 0.38 (0.13–0.59) for CY, and 0.40 (0.13–0.61) for SCY. The correlations of GEBV from ssSNPBLUP increased by an average of 0.08 compared with PED. The average (ranges) of reliabilities were 0.34 (0.15–0.50) for PED, 0.53 (0.28–0.72) for SY, 0.51 (0.27–0.69) for CY, and

Table 5. Correlations and validation reliabilities[1] (in parentheses) of EBV from pedigree-based BLUP (PED) and genomic EBV from single-step (ss)SNPBLUP [residual polygenic variance (k) = 0.10] by groups of genotyped animals for the validation cows (n = 22,594)[2,3]

Trait[4]	PED		SY		CY		SCY	
sta	0.50	(0.50)	0.60	(0.70)	0.59	(0.69)	0.61	(0.72)
cwi	0.27	(0.29)	0.34	(0.43)	0.33	(0.42)	0.34	(0.45)
bde	0.36	(0.36)	0.45	(0.58)	0.44	(0.56)	0.46	(0.60)
ang	0.25	(0.30)	0.30	(0.45)	0.30	(0.44)	0.31	(0.47)
ran	0.38	(0.34)	0.49	(0.58)	0.49	(0.56)	0.51	(0.61)
rls	0.21	(0.21)	0.30	(0.44)	0.29	(0.40)	0.31	(0.46)
fan	0.10	(0.15)	0.13	(0.28)	0.13	(0.27)	0.13	(0.29)
fua	0.32	(0.46)	0.40	(0.72)	0.38	(0.66)	0.41	(0.75)
ruh	0.34	(0.43)	0.41	(0.62)	0.40	(0.59)	0.42	(0.64)
ruw	0.24	(0.26)	0.30	(0.43)	0.30	(0.42)	0.31	(0.47)
usu	0.23	(0.28)	0.31	(0.49)	0.31	(0.48)	0.32	(0.52)
ude	0.48	(0.49)	0.57	(0.71)	0.56	(0.69)	0.58	(0.73)
ftp	0.36	(0.33)	0.45	(0.51)	0.44	(0.49)	0.46	(0.53)
Average	0.31	(0.34)	0.39	(0.53)	0.38	(0.51)	0.40	(0.56)

[1]Validation reliability = the square of correlation divided by the heritability.

[2]All results were calculated by the model that included unknown-parent groups.

[3]PED = pedigree-based BLUP; SY = ssSNPBLUP consisting of sires with classified daughters and young animals; CY = ssSNPBLUP consisting of cows with records and young animals; SCY = ssSNPBLUP consisting of sires with classified daughters, cows with records, and young animals.

[4]sta = stature; cwi = chest width; bde = body depth; ang = angularity; ran = rump angle; rls = rear leg side view; fan = foot angle; fua = fore udder attachment; ruh = rear udder height; ruw = rear udder width; usu = udder support; ude = udder depth; ftp = front teat placement.



Figure 3. Regression coefficients of daughter yield deviations (DYD) from pedigree BLUP in the full data set on EBV from pedigree-based BLUP (PED), and genomic EBV from single-step (ss)SNPBLUP based on groups of genotyped animals in the truncated data set for the validation bulls (n = 762). PED = pedigree-based BLUP; SY = ssSNPBLUP consisting of sires with classified daughters and young animals; CY = ssSNPBLUP consisting of cows with records and young animals; SCY = ssSNPBLUP consisting of sires with classified daughters, cows with records, and young animals; NOUPG and UPG = without unknown-parent group (UGP) and with UPG; 10, 20, and 30 = assumed proportion of polygenic residual variance = 0.10, 0.20, and 0.30; sta = stature; cwi = chest width; bde = body depth; ang = angularity; ran = rump angle; rls = rear leg side view; fan = foot angle; fua = fore udder attachment; ruh = rear udder height; ruw = rear udder width; usu = udder support; ude = udder depth; ftp = front teat placement.

0.56 (0.29–0.75) for SCY. Correlations and reliabilities showed similar trends, with SCY being the highest and CY the lowest among the groups.

Although the number of genotyped cows in the truncated data set was only 6,897, the use of genotyped cows with records slightly improved the predictive ability of GEBV for young animals compared with the use of only genotyped bulls with daughters.

Only bulls with daughters in Japan were included in the analyses. In practice, foreign bulls with international EBV are included according to multiple across-country evaluation (**MACE**). In the multistep genomic evaluation implemented in Japan, the reference population included approximately 6,000 foreign genotyped bulls with MACE EBV. The inclusion of MACE EBV into ssSNPBLUP can increase the reliability of GEBV for young animals. Alkhoder et al. (2022) considered the use of bull MACE EBV for the ssSNPBLUP in German Holsteins and this may have resulted in higher correlations than the ssGBLUP of Tsuruta et al. (2021), which used only US Holsteins national data.

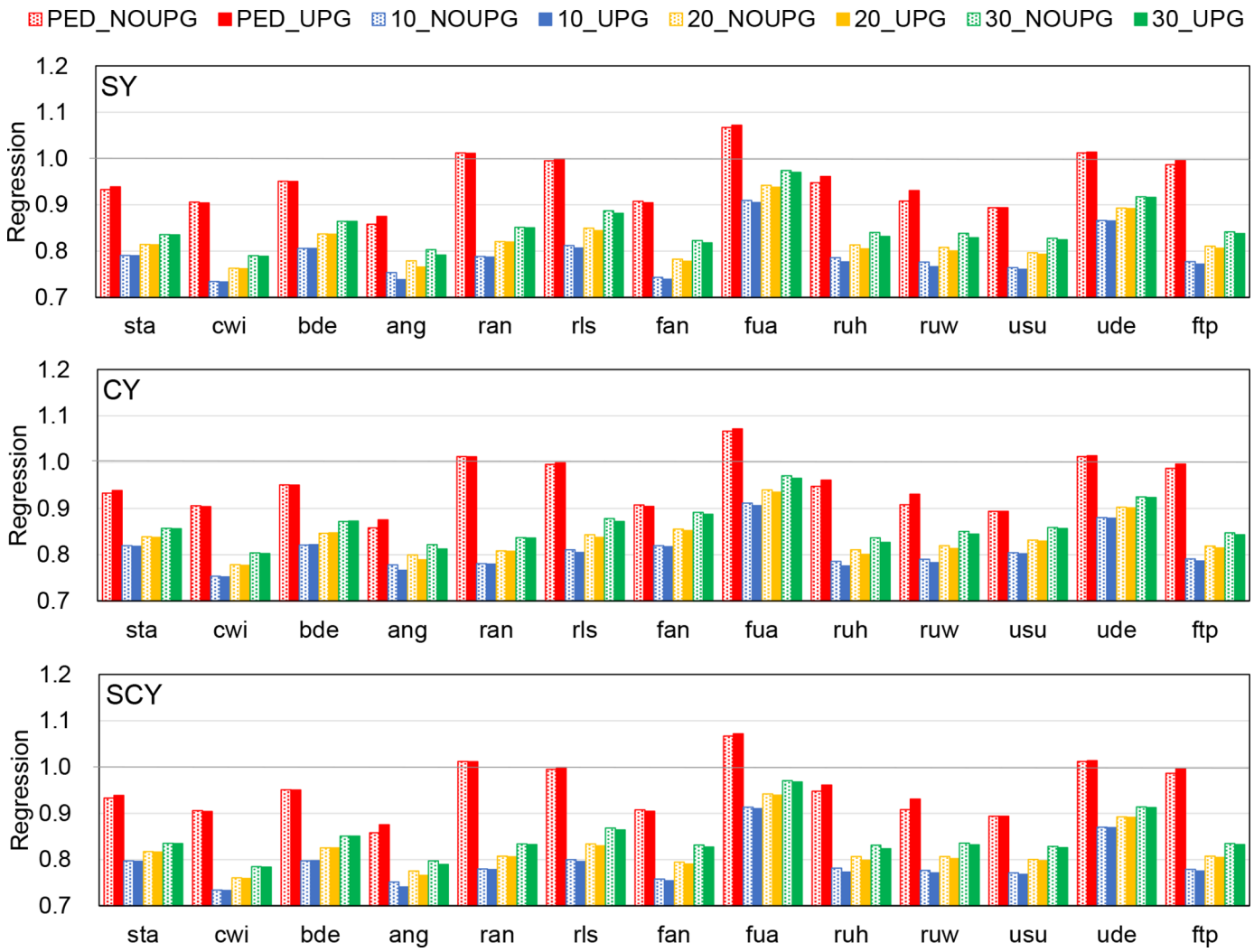


Figure 4. Regression coefficients of adjusted phenotypes (Y_{adj}) from pedigree-based BLUP in the full data set on EBV from pedigree-based BLUP (PED), and genomic EBV from single-step (ss)SNPBLUP based on groups of genotyped animals in the truncated data set for the validation cows (n = 22,594). PED = pedigree-based BLUP; SY = ssSNPBLUP consisting of sires with classified daughters and young animals; CY = ssSNPBLUP consisting of cows with records and young animals; SCY = ssSNPBLUP consisting of sires with classified daughters, cows with records, and young animals; NOUPG and UPG = without unknown-parent group (UGP) and with UPG; 10, 20, and 30 = assumed proportion of polygenic residual variance = 0.10, 0.20, and 0.30; sta = stature; cwi = chest width; bde = body depth; ang = angularity; ran = rump angle; rls = rear leg side view; fan = foot angle; fua = fore udder attachment; ruh = rear udder height; ruw = rear udder width; usu = udder support; ude = udder depth; ftp = front teat placement.

Biases for Validation Bulls and Cows

Regression coefficients of DYD on GEBV and EBV from PED and ssSNPBLUP for the validation bulls are shown in Figure 3. The regression coefficients in PED were closest to 1.0, but most traits in ssSNPBLUP had regression coefficients less than 1.0, implying an overdispersion in the predictions. However, the regression coefficients approached 1.0 as the parameter of residual polygenic variance increased. The regression coefficients were mostly similar, regardless of the use of UPG across the groups of genotyped animals.

Regression coefficients of Y_{adj} on EBV and GEBV from PED and ssSNPBLUP for the validation cows are shown in Figure 4. Similar to the validation bulls, the regression coefficients in ssSNPBLUP approached 1.0 as the parameter of residual polygenic variance increased. The inclusion of UPG in the model and genotyped groups of animals also showed no significant differences in the regression coefficients.

This study included several traits for which increasing the parameter of residual polygenic variance to 0.3 did not result in regression coefficients for validation bulls greater than 0.9. However, increasing the param-

eter of residual polygenic variance reduced the overdispersion of the GEBV for all traits. In the ssSNPBLUP model, it is important to set an appropriate parameter of residual polygenic variance according to the trait to reduce the overdispersion of GEBV in young animals. Along with developing genomic prediction methods, it is also necessary to increase the number of "reference animals," using reliable genetic evaluations or external information such as MACE EBV of foreign bulls to reduce bias in genomic predictions for Japanese Holsteins.

CONCLUSIONS

In a national evaluation of the type traits of Japanese Holstein populations, the ssSNPBLUP model, including UPG, was implemented. The predictive ability of the ssSNPBLUP model was influenced by the group of genotyped animals, and biases were reduced with the increased residual polygenic variance. Thus, when implementing the ssSNPBLUP model for routine genomic evaluation, it is necessary to appropriately set the parameter of residual polygenic variance for each trait to control the overdispersion of GEBV in young animals. The use of UPG did not significantly affect the predictive ability and overdispersion of GEBV in young animals, but differences in the definition of UPG may affect GEBV of individual animals. A further verification of the optimal UPG definition for ssSNPBLUP should be performed in the future.

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